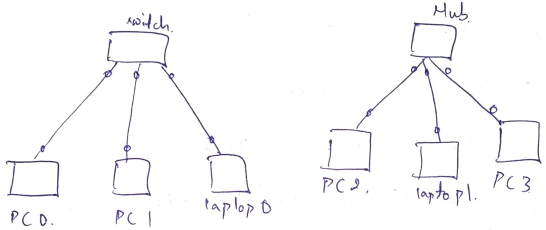
cycle 1.

1. Create a topology and simulate sending a simple PDU from source to destination using hub and switch as connecting devices and demonstrate ping message.

Step 1: Create a simply topology like this.



Step 2: Configure IP address to each devices or hosts as shown below. click on the device

Step 3: Click on, IP configuration. Assign the any class of IP address (class A, class B, class) like this.

Step 4: Repeat the same process to assign IP address for all hosts in the topology.

Step 5: For checking connectivity, select "simple PDU packet" from right side of packet tracer and select source PC and Destination PC, You will get response at right bottom of the packet tracer window.

Step 6: To get the Ping response. Open Desktop of host and select Command Prompt. Run the command shown below.

Output

Command Prompt

Packet Tracer PC Command Line 1.0

PC > ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1 : bytes=32 time<1ms TTL=128

Reply from 192.168.10.1 : bytes=32 time=64ms TTL=128

Reply from 192.168.10.1 : bytes=32 time=63ms TTL=128

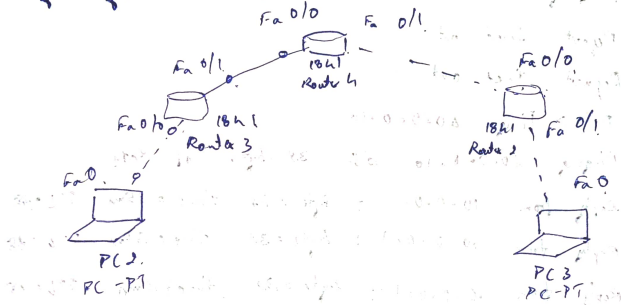
Reply from 192.168.10.1 : bytes=32 time=63ms TTL=128

Reply from 192.168.10.1 : bytes=32 time=63ms TTL=128

10/10 28/7/26

2. Configure IP address to router in packet tracer explore the following messages : Ping responses, destination unreachable request timed out reply

Step 1 : Create a topology like configure IP address & gateway to the host



Step 2 : Configure Interface IP address to router click on Router R1, then click on CLI. You will see like this. " continue with configure " dialog [yes/no] : " give "no" and Press enter. Now you will go to user mode, now give "enable" and press enter : Now you get into the privileged mode, now type, "configure terminal" and press enter to get into global configuration mode

Step 3 : Now configure router interface with IP address & subnet mask then give no. shutdown to make this interface and line protocol up

Step 4: Now lights on all ports become green then
now open cmd of PC and check for ping messa
Destination reachable, request time out & reply

PC > Ping 10.0.0.6

pinging 10.0.0.6 with 32 bytes of data

Request timed out

Request timed out

PC > Ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data

Reply from 10.0.0.1 : byte = 32 time = 25ms TTL =

Reply from 10.0.0.1 : byte = 32 time = 31ms TTL =

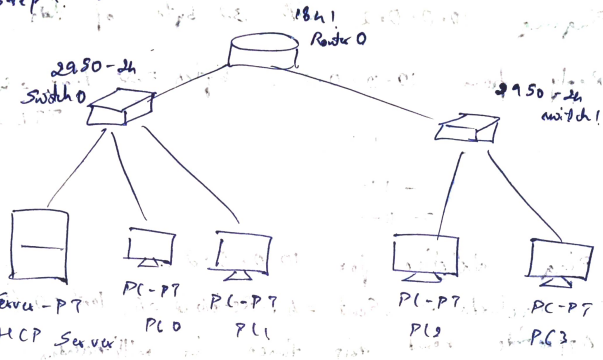
Reply from 10.0.0.1 : byte = 32 time = 31ms TTL =

Reply from 10.0.0.1 : byte = 32 time = 30ms TTL =

05/10/26

3. How to configure DHCP within a LAN in a packet Tracker

Step 1 : Create a LAN.



Step 2 : Click on DHCP. Name → assign gateway in 10.0.0.1

Step 3 : Configure ip address as 10.0.0.2 and sub net mask

Step 4 : Create a server pool for the network On-services- DHCP

Step 5 : Configure router through CLI. To help DHCP you have to add ip address (helper as DHCP server address.

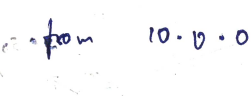
Step 6 : Now select the DHCP option in IP configuration of each PCs. Observe within LAN and outside LAN.

Output:

PC > ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data

Reply from 10.0.0.2 : bytes = 32 time = 1ms TTL = 64



Ping statistics for 10.0.0.2 :

Packets : Sent = 4, Received = 4, Lost = 0 (0% loss)

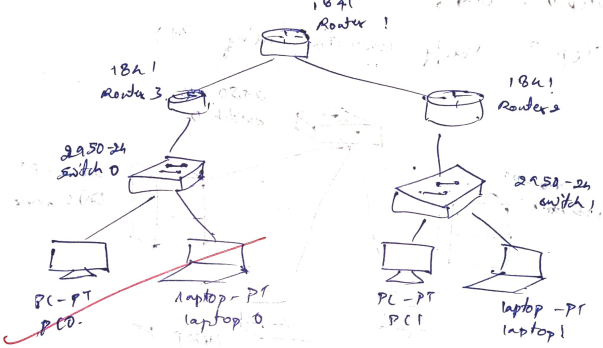
Approximate round trip times in milli-seconds :

Lab 5/10/24

10

4. How to configure static route and default Route to the Routers.

Step 1 : Create a topology with two subnetworks.
Step 2 : Config static route on R2 to route packets
Step 3 : Config default route on R1 and R3
Step 4 : check routing table of R1, R2, R3
Step 5 : Ping 10.0.0.1 to 40.0.0.1 in cmd



Out put

PC > ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:

Reply from 40.0.0.10 byte = 32 time = 1ms TTL = 128

Ping statistics for 40.0.0.10:

Packets sent = 4, received = 4 Lost = 0 (0% loss)

Approximate round trip times in milliseconds

10/10/26

10

5: Configure web server, DNS within a LAN

Step 1: Create a topology

Step 2: Config web server with IP address and DNS

Step 3: Config DNS server to IP address

Step 4: Enable DNS service

Step 5: Enable HTTP service

Step 6: Config host ip address and DNS address

Step 7: Verify connection using CMD

